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<https://swayam.gov.in>https://swayam.gov.in/nc_details/NPTEL

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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction To Machine Learning - IITKGP
(course)



Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

Week 2 ()

- ☐ Lecture 06 : Linear Regression (unit? unit=14&lesson=15)

- ☐ Lecture 07 : Introduction to Decision Trees (unit? unit=14&lesson=16)

Week 2 : Assignment 2 (Revised)

The due date for submitting this assignment has passed.

Due on 2024-08-21, 23:59 IST.

Assignment submitted on 2024-08-21, 18:58 IST

1) 2 points

In a binary classification problem, out of 30 data points 12 belong to class I and 18 belong to class II. What is the entropy of the data set?

- A. 0.97
- B. 0
- C. 1
- D. 0.67

- ☒ A.
- ☐ B.
- ☐ C.
- ☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

A.

2) 2 points

Which of the following properties are characteristics of decision trees?

- A. Low bias
- B. High variance
- C. Lack of smoothness of prediction surfaces
- D. None of the above

☐ Lecture 08 :
Learning
Decision Tree
(unit?
unit=14&lesso
n=17)

☐ Lecture 09 :
Overfitting
(unit?
unit=14&lesso
n=18)

☐ Lecture 10:
Python
Exercise on
Decision Tree
and Linear
Regression
(unit?
unit=14&lesso
n=19)

☐ Lecture 11:
Tutorial - II
(unit?
unit=14&lesso
n=20)

☒ Week 2 :
Lecture
Material (unit?
unit=14&lesso
n=21)

☒ Quiz: Week 2
: Assignment
2 (Revised)
(assessment?
name=153)

☐ Feedback
Form for Week
2 (unit?
unit=14&lesso
n=22)

☐ Assignment 2
Solution (unit?
unit=14&lesso
n=121)

Week 3 ()

Week 4 ()

Week 5 ()

- ☒ A.
☒ B.
☒ C.
☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

- A.
B
C.

3)

Statement: Decision Tree is an unsupervised learning algorithm.

Reason: The splitting criterion uses only the features of the data to calculate the respective measures.

- A. Statement is True. Reason is True.
B. Statement is True. Reason is False.
C. Statement is False. Reason is True.
D. Statement is False. Reason is False.

- ☐ A.
☐ B.
☐ C.
☒ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

D.

4)

In linear regression, our hypothesis is $h_{\theta}(x) = \theta_0 + \theta_1 x$, the training data is given in the table.

x	y
10	5
3	3
6	7
8	6

If the cost function is $J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x_i) - y_i)^2$, where m is no. of training data points.

What is the value of $J(\theta)$ when $\theta = (1, 1)$?

- A. 0
B. 5.75
C. 4.75
D. 6.75

- ☐ A.
☒ B.
☐ C.
☐ D.

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[Week 6 \(\)](#)[Week 7 \(\)](#)[Week 8 \(\)](#)[Assignment
Solution \(\)](#)[Download
Videos \(\)](#)[Problem
Solving
Session -
July 2024 \(\)](#)

Yes, the answer is correct.

Score: 2

Accepted Answers:

B.

5)

2 points

What is a common indicator of overfitting in a decision tree?

- A. The training accuracy is high while the test accuracy is low.
- B. The tree is shallow.
- C. The tree has only a few leaf nodes.
- D. The tree's depth matches the number of attributes in the dataset.

☒ A.

☐ B.

☐ C.

☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

A.

6)

2 points

What is true for Batch Gradient Descent?

- A. In every iteration, model parameters are updated based on one training sample
- B. In every iteration, model parameters are updated based on all training samples
- C. None of the above

☐ A.

☒ B.

☐ C.

Yes, the answer is correct.

Score: 2

Accepted Answers:

B.

Answer Questions 7-8 with the data given below:

Consider the following dataset. We want to build a decision tree classifier to detect whether a tumor is malignant or not using several input features such as age, vaccination, tumor size and tumor site. The target variable is "Malignant" and the other attributes are input features.

Age	Vaccination	Tumor Size	Tumor Site	Malignant
5	1	Small	Shoulder	0
9	1	Small	Knee	0
6	0	Small	Marrow	0
6	1	Medium	Chest	0
7	0	Medium	Shoulder	0
8	1	Large	Shoulder	0
5	1	Large	Liver	0
9	0	Small	Liver	1
8	0	Medium	Shoulder	1
8	0	Medium	Shoulder	1
6	0	Small	Marrow	1
7	0	Small	Chest	1

7) What is the initial entropy of the dataset?

2 points

- A. 0.543
- B. 0.9798
- C. 0.8732
- D. 1

- ☐ A.
- ☒ B.
- ☐ C.
- ☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

B.

8)

2 points

For the dataset in Question 7, what is the information gain of Vaccination (If entropy measure is used to calculate information gain)?

- A. 0.4763
- B. 0.2102
- C. 0.1134
- D. 0.9355

- ☒ A.
- ☐ B.

☐ C.

☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

A.

9)

2 points

Which of the following criteria is typically used for optimizing in linear regression?

- A. Maximizing the number of points touched by the line
- B. Minimizing the number of points touched by the line
- C. Minimizing the sum of squared distance of the line from the points
- D. Minimizing the maximum squared distance of a point from a line

☐ A.

☐ B.

☒ C.

☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

C.

10) The parameters obtained in linear regression

2 points

- A. can take any value in the real space
- B. are strictly integers
- C. always lie in the range [0,1]
- D. can take only non-zero values

☒ A.

☐ B.

☐ C.

☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

A.